BC311 MICROBIAL ENGINEERING (3-1-0)

OBJECTIVE: The microbial engineering provides the knowledge to works on the biological, chemical and engineering aspects of biotechnology, manipulating microbes and developing new uses for microbes.

COURSE CONTENT

UNIT I

Introduction to microbial world and synthesis / formation of different products

UNIT II

Microbial growth: Aerobic and Anaerobic growth phenomena. Different cultivation methods for aerobic and anaerobic cultures. Microbial Growth kinetics and development of growth model, Maintenance Energy – Determination of maintenance coefficient.

UNIT III

Principle and mechanism of media sterilization – Thermal and Membrane filteration; Kinetics of media sterilization, Batch and continuous sterilization of media Air Sterilization – Principles, mechanism and design.

UNIT IV

Different modes of Bioreactors and its basics – Batch, Plug Flow and Continuous cultivation of microorganism. Substrate utilization and product formation kinetics. Discussion of product synthesis kinetics using “Gaden’s Classification and Deindoefer’s classification” of microbial type reactions. Aeration and agitation in microbial fluid. Characteristics of biological fluids: Rheology of fermentation broth.

TEXTBOOK AND REFERENCE BOOKS:

1. Biochemical Engineering by Shuichi Aiba, Arthur E. Humphrey and Nancy F. Millis, *Academic Press: New York*

2. Biochemical Engineering fundamentals by James E. Bailey and David F. Ollis, *McGraw Hill: New York.*

3. Biochemical Engineering : Basuic concepts By Michael L. Shuler, Fikret Kargi, *Prentice Hall.*

4. Process Biotechnology: Theory and Practices by S. N. Mukhopadhyay, *The Energy and Resource Institute (TERI).*

5. Introduction to Biochemical Engineering By D. G. Rao, *Tata McGraw Hill*